

Comprehensive Calibration & Flight Test Routine

Custom ESP32 & ATmega328P Bare-Metal Flight Control System — F330 Frame

● CRITICAL SAFETY RULES (NON-NEGOTIABLE)

- **ALWAYS** remove all propellers during any bench or calibration testing where motors may spin.
- **NEVER** arm the vehicle while anyone is within 2 meters of the propeller arc.
- **ALWAYS** perform isolated ESC calibration and motor direction checks before the first flight.
- Maintain a fully charged LiPo battery; low voltage mid-flight induces system brownouts and crashes.
- A dedicated spotter must be present for all untethered flight operations.
- Conduct all outdoor tests in wide-open environments away from people, obstacles, structures, and power lines.

Note on "Power Cycling": This operation refers to completely disconnecting the high-voltage XT60 battery terminal from the power distribution board, waiting 3 to 5 seconds to allow internal decoupling capacitors to fully drain, and re-engaging the connector to reboot all onboard microcontrollers from a clean power-on-reset state.

Phase 1 — Bare-Metal ESC Calibration (One-Time, Props OFF)

Electronic Speed Controllers (ESCs) must map the discrete pulse width limits generated by your flight controller hardware timers. Because a custom bare-metal ESP32 initialization sequence takes slightly longer to stabilize internal registers than the instant boot time of commercial ESCs, a specific isolation order must be executed to capture the timing limits accurately.

1.1 Hardware Isolation Configurations

To avoid dangerous parasitic cross-core back-feeding and current overdraw from your computing station's USB port, strip down the peripheral bus networks before applying power:

- **Isolate Main Power:** Disconnect the 5V UBEC line from the ESP32 V5 pin.
- **Isolate Coprocessor:** Unplug the UART data connections (TX/RX) and power rails of the ATmega328P supervisor completely to eliminate data-line ghost powering.
- **Isolate Receiver:** Disconnect the FlySky iA6B receiver module entirely.
- **Keep Intact:** Keep the ESC Signal and Ground wires connected to the target ESP32 PWM pins, and ensure the GY-521 IMU remains connected to the 3.3V reference rail.

1.2 Execution Sequence

1. **Flash Temporary Driver:** Ensure your `app_main()` is modified to hardcode a continuous maximum throttle signal (**2000µs** pulse width, or 10% duty cycle on a 13-bit / 50 Hz configuration) instantly upon booting up.
2. **Pre-Boot the Flight Controller:** Plug the ESP32 into your computer via a USB cable. The hardware timer will instantly start screaming the high endpoint signal down the motor lines while the ESCs remain dead.
3. **Engage High-Side Voltage:** Connect your fully charged 12.6V 3S LiPo battery into the main XT60 terminal.
4. **Listen for Signal Capture:** The ESCs will boot, immediately trap the pre-existing high pulse, and emit two short, distinct initialization chimes confirming calibration entry.
5. **Slam to Minimum:** Within the program's scheduled timeout delay, command your code to drop the duty cycle to minimum throttle (**1000µs** pulse width, or 5% duty cycle).
6. **Capture Confirmation:** Listen for a long, steady confirmation note from the ESCs indicating both parameters are written to internal memory. Unplug the XT60 battery, disconnect the USB cable, and restore your normal flight loop code.

Phase 2 — Motor Direction Verification (Props OFF)

Each motor must turn in its mathematically prescribed direction for the standard quadcopter X-frame mixing laws to correctly distribute stability forces.

2.1 Target Rotational Matrix

Motor Position	Target Pin (ESP32)	Required Direction
M1 (Front-Right)	GPIO 13	Counter-Clockwise (CCW)
M2 (Front-Left)	GPIO 25	Clockwise (CW)
M3 (Back-Left)	GPIO 26	Counter-Clockwise (CCW)
M4 (Back-Right)	GPIO 27	Clockwise (CW)

2.2 Procedure

1. Reassemble the standalone flight environment by reconnecting your 5V UBEC power wire, the receiver links, and the ATmega328P UART telemetries.
2. Turn on the FlySky RC transmitter. Power cycle the drone via the battery line and verify the system hits `STATE_DISARMED`.
3. Execute your bare-metal stick arming sequence (Throttle to minimum + Yaw fully right, holding steady for 0.5 seconds).

4. Slowly raise the throttle stick just above the lowest idle threshold (~1100–1150 μ s) until all four motor shafts begin spinning slowly.
5. Observe the rotation from directly above. If any motor spins reversed, power down the vehicle and swap any **TWO of the three physical phase wires** bridging that motor to its respective ESC. Re-test.

Phase 3 — IMU & Sensor Fusion Verification (Props OFF)

3.1 Static Level Baseline

- Place the vehicle on an explicitly level surface confirmed with a spirit level.
- Power up the craft standalone and do not touch, bump, or shake the frame for 5 seconds while your code collects 500 samples to compute the gyro sensor offsets.
- Monitor your live UDP telemetry stream on your computer terminal via: `nc -u 1 4444`
- Verify that the parsed `roll_angle` and `pitch_angle` telemetry values sit tightly within $\pm 0.5^\circ$ of absolute zero.

3.2 Dynamic Orientation Check

Physically tilt the quadcopter by hand to check axis alignment:

- **Pitch Forward (~15°):** Telemetry `pitch_angle` must increase positively.
- **Roll Left:** Telemetry `roll_angle` must increase positively.
- **Roll Right:** Telemetry `roll_angle` must move inversely into negative values.
- *If any response moves backward, check the raw sensor data coordinate mappings (`ax`, `ay`, `az` assignments) relative to your structural mounting orientation inside your flight task configuration.*

3.3 Gyro Drift Stress Test

Leave the drone stationary for 60 consecutive seconds. The total drift across both the roll and pitch tracking registers over the full minute must remain under $\pm 1.0^\circ$. Excessive drifting implies that high-frequency vibration or premature handling corrupted your initialization window.

Phase 4 — Motor Mixing & Control Law Directionality (Props OFF)

This phase verifies that the sign changes in your PID control mixing equations map corrections to the correct corners before trusting the loops in flight.

1. Arm the vehicle and establish a low, uniform bench idle (~1150 μ s).
2. **Roll Right Stick Input:** M1 and M4 must immediately escalate in rotational frequency; M2 and M3 must audibly spin down.
3. **Pitch Forward Stick Input:** M1 and M2 must escalate in speed; M3 and M4 must drop in speed.
4. **Yaw Right Stick Input:** The counter-clockwise motors (M1 and M3) must aggressively accelerate, while the clockwise motors (M2 and M4) spin down.

5. *Correction: If an entire axis reacts inversely to your real control input sticks, invert the sign (+ to -) of that specific PID output variable inside your C-code mixing calculation matrix.*

Phase 5 — Tethered Hover & Balance Testing (Props ON)

This stage transitions your bare-metal code into handling physical aerodynamic thrust forces under absolute mechanical constraint.

- **Propeller Attachment:** Securely tighten your propellers onto the shafts. Ensure CCW-designated props sit perfectly on CCW-spinning motors (M1, M3) and CW-designated props sit on CW motors (M2, M4).
- **The Structural Tether:** Utilize high-tensile nylon zip-ties or heavy ropes to loosely tether all four arms of the F330 frame directly to a weighted ground structure. Allow the craft roughly 5 to 10 centimeters of free upward travel, but eliminate any possibility of rolling or drifting away.
- **Thrust Balance Verification:** Step back to a safe boundary, arm the quadcopter, and smoothly feed throttle until the vehicle begins floating against its tethers. If the frame leans heavily towards a specific corner, adjust the position of your 3S battery pack to align the physical Center of Gravity (CoG) directly underneath the center intersection of the frame axes.

Phase 6 — PID Discrete Domain Control Tuning

Because your real-time `task_flight` controller code runs a discrete time-step execution loop at a fixed frequency ($250\text{ Hz} \rightarrow T_s = 0.004s$), traditional continuous equations do not directly apply. Your embedded registers must handle discrete coefficients:

```
// Bare-Metal Discrete Mapping for PID Accumulation
p->integral += error * LOOP_TIME_S; // Discrete Integral Integration
float D = p->Kd * (error - p->prev_err) / LOOP_TIME_S; // Discrete Derivative
Derivative
```

6.1 Tuning Methodology (Modified Ziegler-Nichols)

1. **Isolate Parameters:** Zero out the integral and derivative gains ($K_i = 0$, $K_d = 0$). Initialize your proportional gain register at a minimal baseline ($K_p = 0.1$).
2. **Locate Ultimate Gain:** Step up K_p sequentially by increments of 0.1 until the tethered vehicle enters a continuous, sustained oscillation of uniform amplitude when disturbed. Record this critical boundary value as your Ultimate Gain (K_u).
3. **Measure Oscillation Period:** Extract the time delta between two matching telemetry wave peaks from your logged UDP stream data. Record this period value as T_u (in seconds).
4. **Compute Coefficients:** Calculate your operational gains using the standard multipliers:
 - Proportional Gain: $K_p = 0.60 \times K_u$
 - Integral Gain: $K_i = 2 \times K_p / T_u$

◦ Derivative Gain: $K_d = K_p \times T_u / 8$

5. **Routh-Hurwitz Stability Check:** To guarantee long-term control loop stability, your computed gains must satisfy the strict physical constraint: $K_d \times K_p > J \times K_i$ (where $J \approx 0.003 \text{ kg}\cdot\text{m}^2$ represents the approximate inertia matrix of your F330 quadcopter frame). If this equation is violated, manually damp down your K_i register until stability is restored.

Phase 7 — First Free Flight Execution Profile

Only transition to untethered flight operations once your vehicle can maintain a solid, level, stable hover while tied down to the testing bench for at least 30 uninterrupted seconds.

7.1 Staging Environment & Takeoff

- Move to an isolated outdoor arena with at least 10m × 10m of clear clearance. Ensure ambient wind measurements remain under 15 km/h.
- Place the drone down on flat terrain, orienting the rear frame elements directly facing the pilot position.
- Power up your radio transmitter, engage the 12.6V battery connection, step back a minimum of 3 meters, and smoothly apply throttle.
- Pop the vehicle above the immediate ground-effect pocket up to an altitude of 0.3 to 0.5 meters. Hold this position for 15 seconds to evaluate structural leveling behavior.

7.2 Critical Emergency Failsafe Validation

While maintaining a secure hover at roughly 1 meter over a soft grass buffer zone, have your spotter physically turn off your FlySky radio transmitter power switch while you maintain your stance clear of the drone.

Required Response: The bare-metal flight loop on Core 1 must immediately register the packet frame timeout gap, trip into STATE_FAILSAFE within **500 ms**, instantly drop all motor PWM signals to a zero-throttle duty cycle (**1000μs**), and disarm. This validates the hardwired software protection mechanism against dangerous runaway scenarios.

Phase 8 — Coprocessor Battery Telemetry & System Health Matrix

The secondary ATmega328P coprocessor continuously evaluates system electrical properties via an analog read of pin ADC0 (A0) tied across your 10kΩ/4.7kΩ voltage divider network. Its internal digital input buffer register is disabled via `DIDR0 |= (1 << ADC0D)` to mitigate intermediate logic switching noises.

8.1 Voltage Management Thresholds (3S LiPo)

Cell Voltage	Total Pack Voltage	Required System Operational Status
4.20 V	12.60 V	Nominal capacity. Fully charged and safe for takeoff initialization.
3.80 V	11.40 V	Nominal flight range. Safe operating zone. Storage threshold.
3.50 V	10.50 V	Low Voltage Limit. ATmega coprocessor activates rapid piezo alert beeps. Land immediately.
3.40 V	10.20 V	Critical Minimum. Severe risk of ESC brownout, logic lockups, and total structural failure.

Phase 9 — Structural Post-Flight Verification Checklist

Execute this sequence comprehensively within 30 seconds of every single flight landing session before packing your equipment away:

- [] **Isolate Power:** Disengage the high-voltage XT60 battery connector immediately upon touchdown.
- [] **Propeller Integrity:** Inspect all four composite blades for stress fracturing, hairline stress cracks, edge micro-chipping, or hub delamination. Replace any suspect components immediately.
- [] **Fastener Torque:** Check all motor structural mounting screws; high-frequency motor vibrations naturally back out unthreaded hardware over time.
- [] **Thermal Assessment:** Physically check the temperature of all four brushless motor bell housings and ESC casings. Excessively hot components indicate bad PID loop filtering or motor mixing imbalances.
- [] **Data Integrity Check:** Inspect the structural seat of your I2C jumper connections (SDA/SCL) and the UART telemetry lines linking your ESP32 to your ATmega328P supervisor to ensure nothing shifted out of place during high-vibration flight maneuvers.